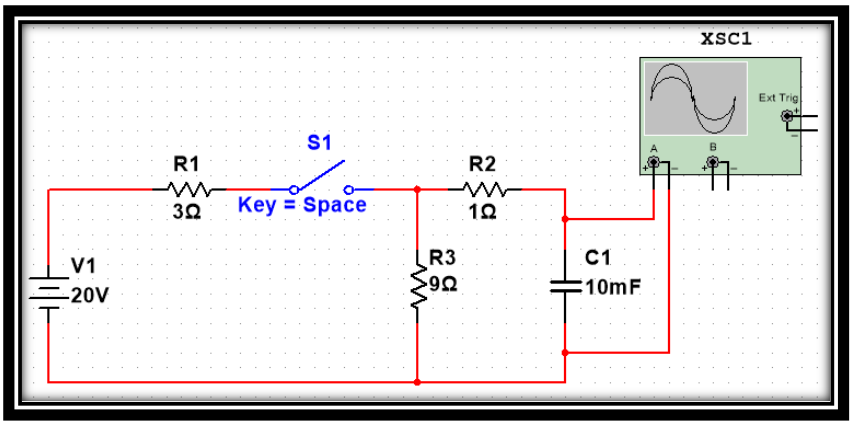
Nama Anggota : 1. Erika Dwi Jayanti (211331006)
2. Shalula Arbani Dynastillah (211331027)
3. Tsaniya Faridah Maulida (211331031)
Kelas : 1A – TTE

PRAKTIKUM RANGKAIAN LISTRIK 2

(RANGKAIAN TRANSIEN RC)

1) **Vs = 20 volt**
   R1 = 3Ω
   R2 = 1 Ω
   R3 = 9 Ω
   C = 10 mF



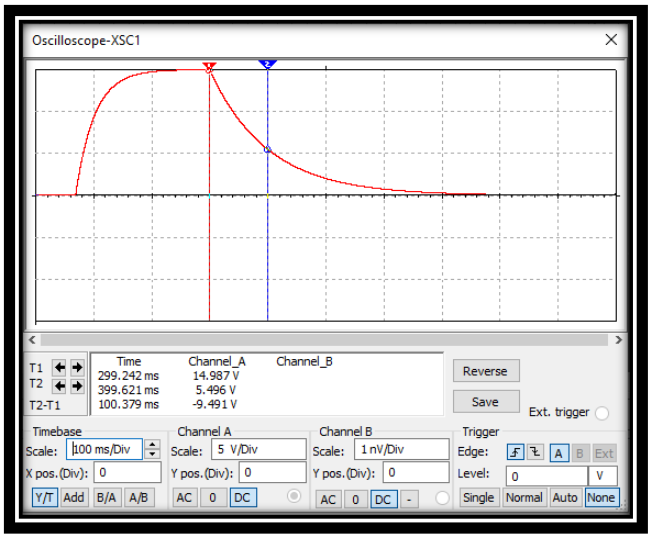
Gambar 1. Gambar Rangkaian Ke-1

$\tau = R_{Eq}C = (10)\ (10\times10^{-3}) = 100$ ms

$2\tau = 2R_{Eq}C = (2)\ (10)\ (10\times10^{-3}) = 200$ ms

$3\tau = 3R_{Eq}C = (3)\ (10)\ (10\times10^{-3}) = 300$ ms

a) Saat $t = \tau$



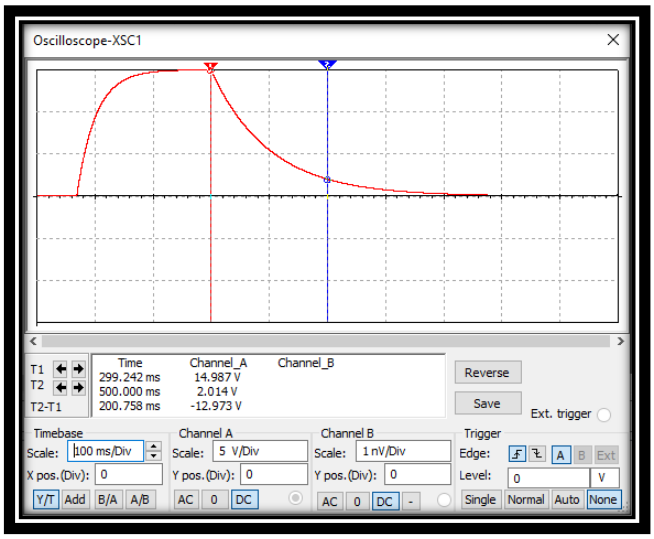
Gambar 2. Display Oscilloscope Saat t = τ

$V_{(t)}$ = 5,496 V

V_0 = 14,987 V

$$t = \frac{V_{(t)}}{V_0} = \frac{5{,}496\ V}{14{,}987\ V} = 0{,}366717$$

b) Saat $t = 2\tau$



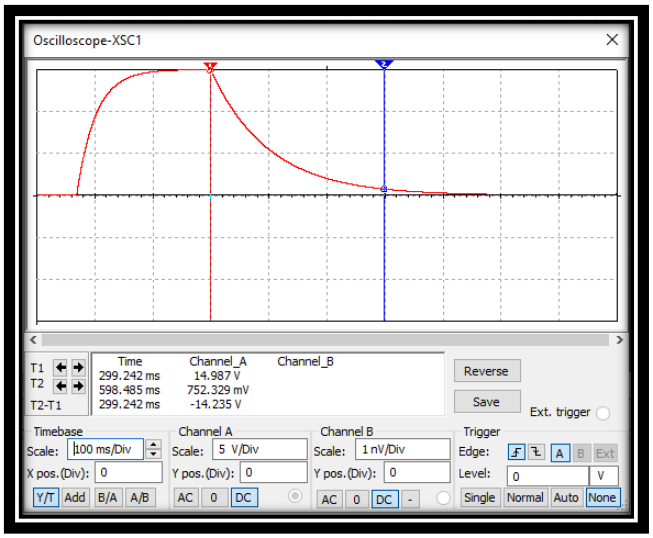
Gambar 3. Display Oscilloscope Saat t = 2τ

$V_{(t)}$ = 2,014 V

V_0 = 14,987 V

$$t = \frac{V_{(t)}}{V_0} = \frac{2{,}014\ V}{14{,}987\ V} = 0{,}134383$$

c) Saat $t = 3\tau$



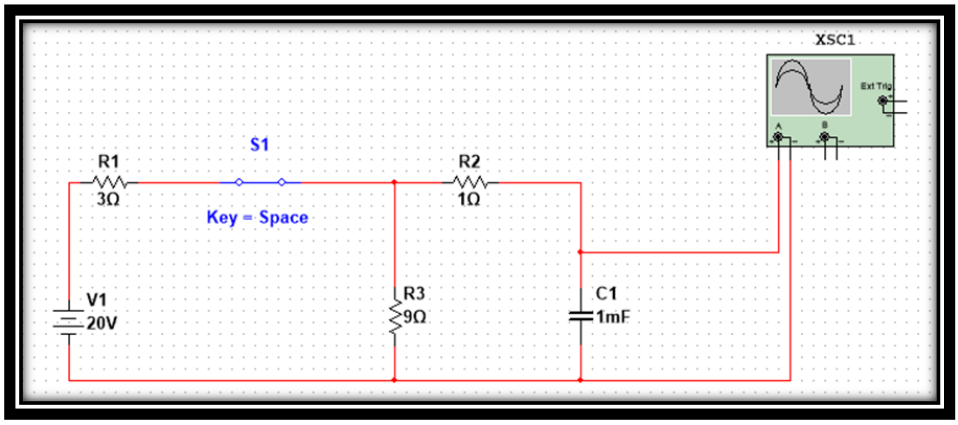
Gambar 4. Display Oscilloscope Saat t = 3τ

$V_{(t)}$ = 0,752329 V

V_0 = 14,987 V

$$t = \frac{V_{(t)}}{V_0} = \frac{0{,}752329\ V}{14{,}987\ V} = 0{,}050198$$

2) **Vs = 20 volt**
R1 = 3Ω
R2 = 1 Ω
R3 = 9 Ω
C = 1 mF



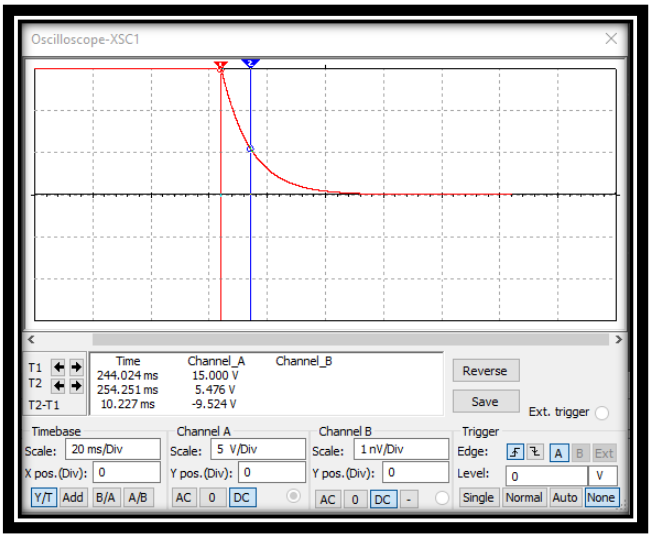
Gambar 5. Gambar Rangkaian Ke-2

$\tau = R_{Eq}C = (10)\ (1\times10^{-3}) = 10$ ms

$2\tau = 2R_{Eq}C = (2)\ (10)\ (1\times10^{-3}) = 20$ ms

$3\tau = 3R_{Eq}C = (3)\ (10)\ (1\times10^{-3}) = 30$ ms

d) Saat $t = \tau$



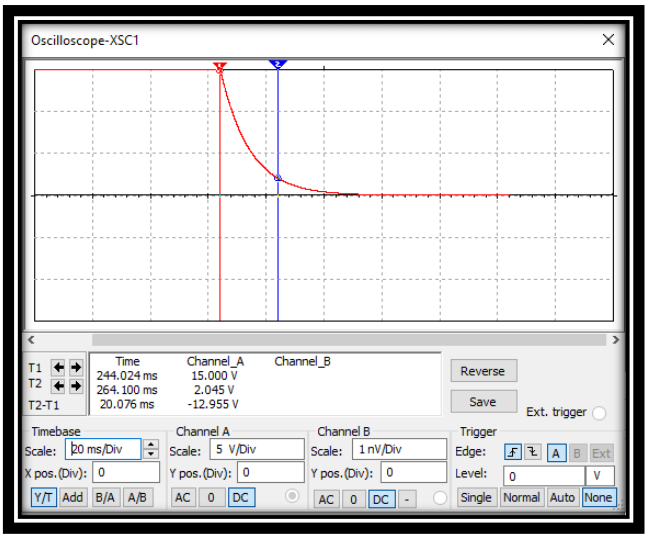
Gambar 6. Display Oscilloscope Saat $t = \tau$

$V_{(t)} = 5{,}476$ V

$V_0 = 15$ V

$$t = \frac{V_{(t)}}{V_0} = \frac{5{,}476\ V}{15\ V} = 0{,}365066$$

e) Saat $t = 2\tau$



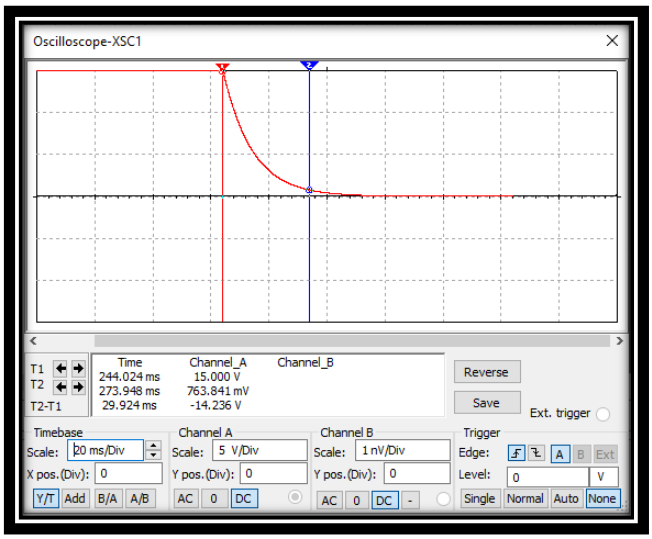
Gambar 7. Display Oscilloscope Saat t = 2τ

$V_{(t)}$ = 2,045 V

V_0 = 15 V

$$t = \frac{V_{(t)}}{V_0} = \frac{2{,}045\ V}{15\ V} = 0{,}136333$$

f) Saat $t = 3\tau$



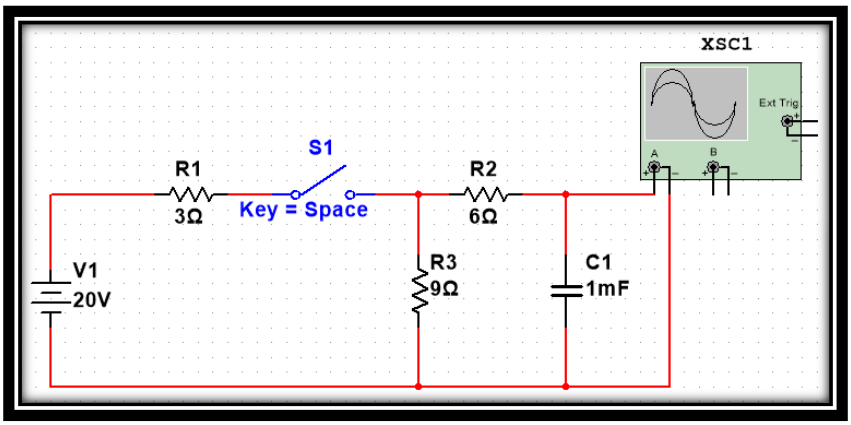
Gambar 8. Display Oscilloscope Saat t = 3τ

$V_{(t)}$ = 0,763841 V

$V_0 = 15$ V

$t = \frac{V_{(t)}}{V_0} = \frac{0{,}763841\ V}{15\ V} = 0{,}050922$

3) **Vs = 20 volt**
R1 = 3Ω
R2 = 6 Ω
R3 = 9 Ω
C = 1 mF



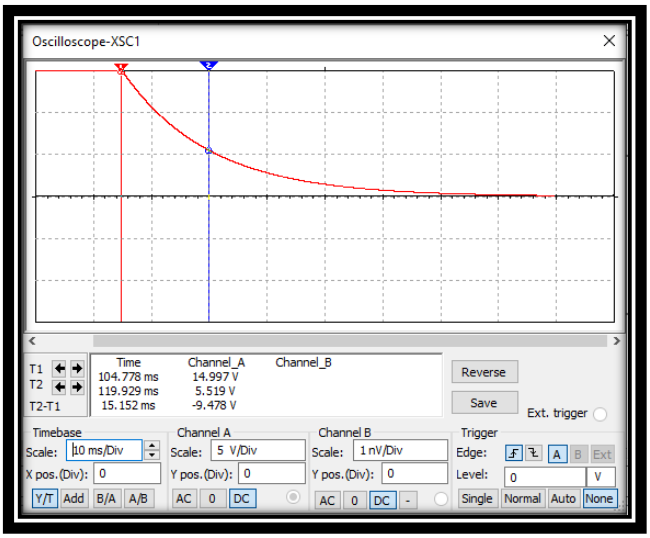
Gambar 9. Gambar Rangkaian Ke-3

$\tau = R_{Eq}C = (15)\ (1\times10^{-3}) = 15$ ms

$2\tau = 2R_{Eq}C = (2)\ (15)\ (1\times10^{-3}) = 30$ ms

$3\tau = 3R_{Eq}C = (3)\ (15)\ (1\times10^{-3}) = 45$ ms

a) Saat $t = \tau$



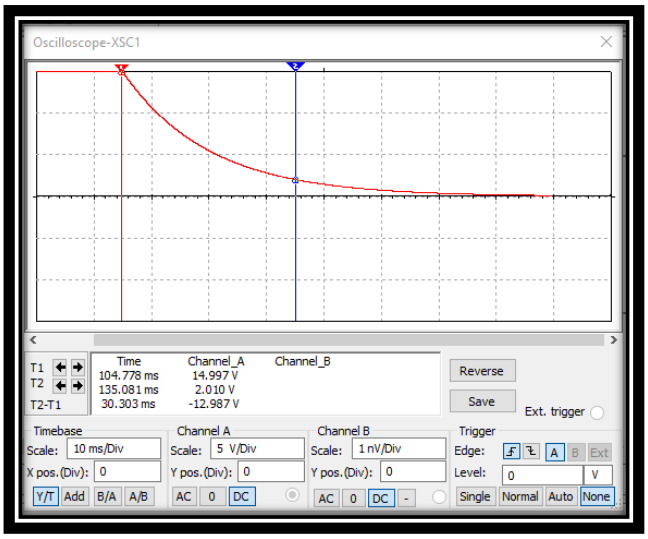
Gambar 10. Gambar Oscilloscope Saat $t = \tau$

$V_{(t)} = 5{,}519$ V

$V_0 = 14{,}997$ V

$$t = \frac{V_{(t)}}{V_0} = \frac{5{,}519\ V}{14{,}997\ V} = 0{,}368006$$

b) Saat $t = 2\tau$



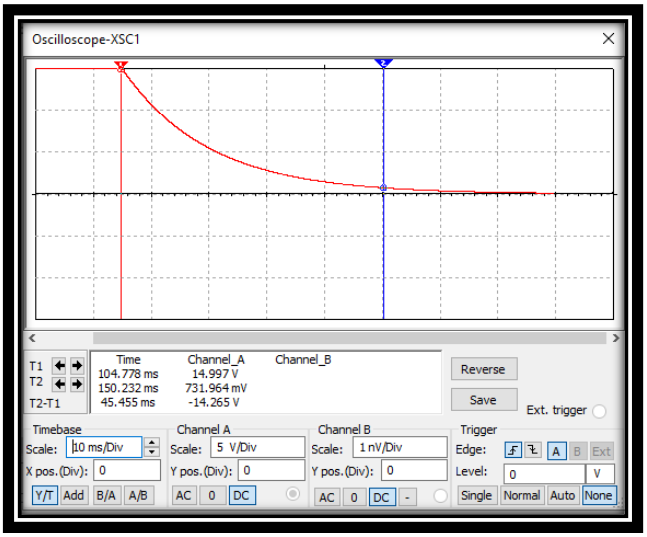
Gambar 21. Display Oscilloscope Saat t = 2τ

$V_{(t)}$ = 2,010 V

V_0 = 14,997 V

$$t = \frac{V_{(t)}}{V_0} = \frac{2{,}010\ V}{14{,}997\ V} = 0{,}134026$$

c) Saat $t = 3\tau$



Gambar 3. Display Oscilloscope Saat t = 3τ

$V_{(t)}$ = 0,731964 V

V_0 = 14,997 V

$$t = \frac{V_{(t)}}{V_0} = \frac{0{,}731964\,V}{14{,}997\,V} = 0{,}048807$$